Karthikheyaa Kurra

karth@udel.edu | GitHub | Portfolio | LinkedIn

Education

University of Delaware — Newark, DE
M.S. Data Science, GPA 3.80/4.00 — *Aug 2024 – May 2026*

- **Coursework:** Probability Theory & Applications, Mathematical Techniques for Data Science (Optimization), Applied Multivariate Statistics, Machine Learning, Algorithm Design & Analysis, Research Software Engineering

Sathyabama Institute of Science & Technology — Chennai, India
B.E. Computer Science & Engineering — *Oct 2020 – May 2024*

Experience

Research Associate, Scientific Machine Learning — Jun 2026 – Present
Department of Mathematical Sciences, University of Delaware (Advisor: Dr. Ke Chen) — *Newark, DE*

- Writing a paper on hybrid PDE solvers that combine classical methods and neural networks to solve a nonlinear elliptic PDE in a non-iterative manner through learned surrogates and by leveraging hierarchical structure.
- Simulated high-precision ground-truth generation via Chebyshev spectral discretizations and Newton iteration, automated dataset-validation suites, and reproducible PyTorch training and evaluation pipelines.
- Investigated architecture through ablation studies covering network depth, normalization, and physics-informed residual losses with adaptive loss balancing, alongside computing the complexity and wall-clock time of inference pipeline.
- Built and Maintained a codebase by writing good code, unit tests, smoke tests and made weekly reports to track research progress.

Research Assistant, Scientific Computing — Jun 2025 – Jun 2026
Department of Mathematical Sciences, University of Delaware (Advisor: Dr. Ke Chen) — *Newark, DE*

- Built a hierarchical Poincaré–Steklov (HPS) numerical solver for linear elliptic PDEs: Chebyshev spectral grids on leaf boxes, local Dirichlet-to-Neumann maps, and Schur-complement merges up the tree.
- Verified the solver on manufactured solutions (Laplace, Poisson, Helmholtz) and wrote notes documenting the leaf operators, merge step, and scaling of the hierarchy.

Teaching Assistant, Machine Learning — Jan 2024 – Apr 2024
Sathyabama Institute of Science & Technology — *Chennai, India*

- Graded assignments, midterms and homework; prepared lecture slides and supplementary notes for the undergraduate machine learning course.
- Led review sessions and tutorials on mathematics for machine learning, covering lecture material on matrix algebra and multivariable calculus.

Research Intern, Computer Vision — Mar 2023 – Jun 2023
Indian Institute of Technology, Hyderabad — *Hyderabad, India*

- Case study applying supervised contrastive learning (Khosla et al., 2020) to facial expression recognition, comparing SupCon loss against a standard cross-entropy baseline in PyTorch with face detection and data augmentation.
- Analyzed learned embeddings with t-SNE to diagnose class separation issues and delivered a reproducible codebase to the lab.

Summer Intern — Jun 2022 – Jul 2022
The University of Texas at Dallas — *Dallas, TX*

- Intensive summer program in machine learning and deep reinforcement learning: lectures, labs, and hands-on model building from first principles.

Technical Skills

Languages: Python, MATLAB, R, SQL, Bash, LaTeX
Scientific ML: PyTorch, NumPy/SciPy, Operator Learning, Physics-Informed Neural Networks (PINNs)
Tools: Git, Docker, Linux, HPC job scripting, VS Code, Cursor

Reading Groups

Scientific Machine Learning Reading Group — 2025 – Present
University of Delaware (Advisor: Dr. Ke Chen) — *Newark, DE*

- Weekly seminar with rotating student presentations of research literature; topics include Computational Optimal Transport (Peyré & Cuturi), Equivariant Neural Networks, and One-Shot Learning.